



Key Selection Criteria

Buyer should evaluate the AOI machine before making a purchase rather than going after vendor-specific marketing claims

RANK	FEATURE TO EVALUATE	WHY IT MATTERS	WHAT BUYERS SHOULD ASK
1	Inspection Accuracy and Defect Coverage	The primary purpose of AOI is reliable defect detection with minimal escapes. The machine should detect solder defects, component placement issues, polarity, bridges, tombstones, lifted leads, missing parts, wrong components, and contamination	Which defects are detected? Are they inspected in both 2D and 3D?
2	False Call Rate (False Rejects)	A machine that generates excessive false alarms wastes operator time and reduces productivity. Most premium AOIs now emphasise AI-based reduction of false calls	What is the false call rate? How is it measured? Any customer references?
3	3D Inspection Technology	Modern electronics increasingly require true 3D measurement rather than only image comparison. Structured light, multiple projectors, and 3D height measurement improve detection of solder volume, lifted components and coplanarity	Is it true 3D? Structured light? Number of projectors? Maximum measurable height?
4	AI-Assisted Programming and Inspection	AI reduces programming time, automatically recognises components, optimises inspection parameters and improves defect classification. This significantly reduces engineering effort	Does AI create inspection programs automatically? Does it learn from operator feedback?
5	Inspection Speed (Throughput)	AOI should not become the bottleneck of the SMT line. Speed depends on camera resolution, field of view (FOV), motion control, and image processing capability	Inspection speed (cm ² /sec or sec/FOV)? Maximum UPH?
6	Camera Resolution And Optical System	Better cameras and optics improve inspection of fine-pitch ICs, tiny passives and solder joints. Telecentric lenses and multiple cameras reduce distortion	Camera megapixels? Lens type? Resolution (5µm, 7µm, 10µm)?
7	Programming Flexibility And Changeover Time	High-mix EMS manufacturers require fast product changeovers. Offline programming, CAD import, and automatic program generation reduce downtime	Does it support CAD/Gerber import? Offline programming? Automatic library generation?
8	MES/SPC/Industry 4.0 Connectivity	Inspection data should integrate with MES, SPC, and traceability systems for process improvement and quality analytics	Supports MES? IPC-CFX? Hermes? SME-MA? SPC reports? Barcode/QR traceability?
9	Capability For Future Components	Electronics continue shrinking. Buyers should ensure support for 0201/01005 components, advanced packages, and large PCBs	Minimum component size? Maximum PCB size? Future upgrade path?
10	Ease Of Maintenance And Remote Support	Fast diagnostics, remote programming, software updates, and centralised management reduce machine downtime and service costs	Remote support? Predictive diagnostics? Multi-machine management?

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Saki, Magic ray

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Brand(s) Supplied: Yamaha, Magic Ray, Omron

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Brand(s) Supplied: Vitrox

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Brand(s) Supplied: Sintek, Quick